

The Retained-Distinction Tensor Field Theory

Supplementary Material: the full derivation, root to curvature, as one continuous machine-checked thread over $\mathbb{Q}$

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Preprint notice. This supplement is a preprint and has not yet undergone formal peer review. Every step below tagged [Th.coqc] is independently confirmable by recompiling the delivered Coq sources [20] — whose type theory is the Calculus of Constructions of Coquand and Huet [6] — under coqc 8.20.1 (exit status 0, **Print Assumptions** = *Closed under the global context*). The reader is invited to audit rather than believe.

How to read this supplement. This is one continuous logical thread. It starts from a single root — the retained difference (RD), i.e. distinguishability — and never leaves the rationals $\mathbb{Q}$. Each step gives (i) the *statement*, (ii) the *Coq theorem / file* that closes it, (iii) its *tier tag*, and (iv) one line of *why* it follows from the step before. The framing invariant is fixed throughout: **the discrete tensor PDE over $\mathbb{Q}$ IS the theory**; the continuum is never a target, gate, or aspiration — it is a [Refused] non-readout (§7).

1 The root: retained difference

Step 0 — RD is the only primitive. *Statement.* The single primitive is the *retained difference* δ_R : whether two states are distinguished *and* that distinction is kept. Everything downstream is a readout of retained differences; no continuum, no real number, and no dimensionful constant is assumed prior to a readout. *Tier.* [Dr] (the programme postulate; every object it generates is then tiered on its own). *Why.* A theory that only ever reads finite retained distinctions can be carried entirely in $\mathbb{Q}$, which is what makes every later step machine-checkable and axiom-free.

Step 1 — distinctions become graph weights. *Statement.* Retained differences between nodes i, j are recorded as non-negative weights w_{ij} on the edges of a graph G ; a node’s total retained distinction is its weighted degree $d_i = \sum_j w_{ij}$. *Tier.* [Dr] over [Th.coqc] objects. *Why.* A weight is just a kept difference given a name; the graph is the minimal object on which “how much is distinguished, and from what” is a finite readout.

2 The root operator: the retained-distinction Laplacian

Step 2 — $L_R = D_W - W$ is forced. *Statement.* The operator that reads “net retained difference at a node relative to its neighbours” is the graph Laplacian [16, 12]

$$L_R = D_W - W, \quad (L_R \phi)_i = \sum_j w_{ij} (\phi_i - \phi_j),$$

with $D_W = \text{diag}(d_i)$. On a 1D chain of unit weights this is exactly the *second finite difference* $(L_R \phi)_j = -(\phi_{j+1} - 2\phi_j + \phi_{j-1})$. *Coq.* core URCF_RD.All; second-difference identity `riemann_fd = w(S(S j)) - 2 w(S j) + w j` in `InfoDiscreteRiemannCurvature.v`. *Tier.* [Th.coqc]. *Why.* $L_R = D_W - W$ is the unique symmetric, difference-only operator that annihilates constants (no distinction $\Rightarrow$ no readout) — it is the discrete spatial-geometry axis, taken as an exact difference, never as ∂_{xx} in a limit.

3 The spine PDE

Step 3 — the master equation. *Statement.* Let Φ be the retained-distinction field over the nodes (tensor-valued once an internal index space is attached). Balancing inertia, dissipation and the geometric coupling gives the one master equation

$$M \partial_{tt} \Phi + D \partial_t \Phi - K \cdot L_R \cdot \Phi + \nabla V(\Phi) = J - \eta$$

with M inertia, D dissipation, K coupling, V self-interaction, J source, and η the finite information-cost readout/back-reaction in the sense of Landauer [15] (an it-from-bit lineage [23]). Setting $V \equiv 0$, $J = \eta$ leaves the *spine* $M \partial_{tt} \Phi + D \partial_t \Phi + K L_R \Phi = 0$. *Coq.* core InfoSchrodinger / InfoDecoherence / InfoTelegraph. *Tier.* spine identity [Th.coqc]; the tensor-PDE envelope reading [Dr]. *Why.* The three named readouts are the three pieces of *one* equation: $M \partial_{tt}$ alone $\rightarrow$ unitary oscillation (wave/quantum, the Schrödinger [18] / Klein–Gordon [13, 9] channel), $D \partial_t$ alone $\rightarrow$ dissipative decay (diffusion), and their sum on $L_R \rightarrow$ the telegraph equation [21, 10] of Step 5.

Step 4 — the actual update (no continuum ∂). *Statement.* On an L_R -eigenmode λ , ∂_{tt} and ∂_t are finite differences; the field advances by the two-register

(position/velocity) update in the Störmer–Verlet lineage [19, 22]

$$\begin{aligned} X[n+1] &= X[n] + \Delta\theta_R V[n], \\ V[n+1] &= V[n] + \Delta\theta_R (K L_R \Phi - D V[n] \\ &\quad - \nabla V(\Phi) + J - \eta)/M, \end{aligned}$$

with n the tick counter and $\Delta\theta_R$ the retained-distinction phase increment (*not* physical dt ; Clock Registry, main text). *Coq.* `decay-ratio / backward-Euler brick InfoDecayRatioFromTauC.v`. *Tier.* [Th_coqc] (the continuum $\exp(-\Gamma_{\text{damp}} t)$, with damping rate $\Gamma_{\text{damp}} = D/(2M)$, is I2, [Refused]). *Why.* The update is the spine *as arithmetic over* $\mathbb{Q}$: every quantity is a finite readout and no limit is ever taken — the discrete dynamics is primary, not an approximation to a smooth one.

Step 4' — the same spine as a stationary action.

Statement. In the Euler–Lagrange first-variation lineage [7, 14], over one node-window $p \rightarrow m \rightarrow q$ the discrete window action

$$A_{\text{win}}(m) = M|q - m|^2 + M|m - p|^2 - h^2 K \text{gform}(m, m)$$

is stationary, $\partial A_{\text{win}}/\partial m = 0$, exactly on the update of Step 4. *Coq.* `InfoActionStationarity.v` (with `InfoGeodesicActionStationarity` giving $M \text{acc} = \frac{1}{2}\gamma \text{vel}^2$, the exact discretisation of $M\ddot{x} = -\frac{1}{2}M'\dot{x}^2$). *Tier.* [Th_coqc]. *Why.* Stationarity is proved as an exact ring identity, not a variation-in-the-limit, so the Euler–Lagrange envelope is a $\mathbb{Q}$ -readout on equal footing with the update.

4 Telegraph form and mass

Step 5 — the spine IS the telegraph equation; $M = 1/2\tau_c$. *Statement.* The spine $M\partial_{tt}\Phi + D\partial_t\Phi + K L_R \Phi = 0$ is *literally* the telegraph equation, and matching the damping to the retained-distinction correlation time τ_c fixes the mass

$$M = \frac{1}{2\tau_c}.$$

The equation is exactly the telegraph equation of Thomson and Heaviside [21, 10]. *Coq.* `core InfoTelegraph`. *Tier.* [Th_coqc] (mass identity exact over $\mathbb{Q}$). *Why.* Mass is not inserted — it is *read out* of the dissipative time-scale of the same equation, closing “what is inertia” in terms of the RD root alone.

5 The discriminant crossover

Step 6 — one algebraic knife-edge splits the regimes. *Statement.* The temporal characteristic on an L_R -eigenmode λ is $p_{\text{char}} = M\omega^2 + D\omega + K\lambda$, with discriminant and crossover

$$\text{disc} = D^2 - 4MK\lambda, \quad \lambda_c = \frac{D^2}{4MK},$$

and the exact completion of the square $4Mp_{\text{char}} = (2M\omega + D)^2 - \text{disc}$. *Coq.* `InfoTelegraphCrossover.v`: `char_complete_square`, `critical_disc_zero`, `underdamped_iff_above_crit`. *Tier.* [Th_coqc]. *Why.* The single sign of disc decides everything below, and it is fixed by an exact algebraic identity, not a numerical threshold.

Step 6a — under-damped = quantum.

Statement. For $\text{disc} < 0$, $p_{\text{char}} > 0$ for every real ω : no real frequency root, hence oscillatory (wave/quantum, the `InfoSchrodinger` side). *Coq.* `underdamped_positive`. *Tier.* [Th_coqc]. *Why.* A strictly positive characteristic has only complex roots — looping, not settling.

Step 6b — over-damped = classical.

Statement. For $\text{disc} \geq 0$, $p_{\text{char}} \leq 0$ at $\omega = -D/2M$: a real decay root exists, hence pure decay (diffusion/classical, the `InfoDecoherence` side). *Coq.* `overdamped_min_nonpositive`. *Tier.* [Th_coqc]. *Why.* A real root is a settling mode — the classical branch is where distinction relaxes.

Step 6c — witnesses.

Statement. $(M, D, K, \lambda) = (1, 1, 1, 1) \Rightarrow \text{disc} = -3 < 0$ (under-damped/quantum); $(1, 4, 1, 1) \Rightarrow \text{disc} = 12 > 0$ (over-damped/classical). *Tier.* [finite_diagnostic]. *Why.* Both regimes are actually realised over $\mathbb{Q}$, so the crossover is non-vacuous.

Step 6d — the crossover IS the horizon = agency knife-edge.

Statement. The telegraph disc is the core `InfoFrontier.spine_discr`, and λ_c is `spine_lambda_c`; the oscillatory (quantum) regime is exactly *above* the horizon. *Coq.* `InfoTelegraphHorizonUnification.v`: `disc_is_spine_discr`, `lam_c_is_spine_lambda_c`, `quantum_iff_above_horizon`, `oscillatory_above_horizon`. *Tier.* [Th_coqc]; reading [Dr]. *Why.* One knife-edge unifies quantum $\leftrightarrow$ classical (dynamics), the black-hole horizon (GR), and the agency threshold: looping above, settling at or below.

6 The discrete curvature tensor

Here $K \cdot L_R$ (Step 2–3) is unfolded into a full discrete curvature tensor, still entirely over $\mathbb{Q}$, still division-free where it matters.

Step 7 — Christoffel = first difference (and $\neq$ curvature).

Statement. The connection is the discrete Christoffel symbol [5], the first finite difference `christoffel_fd w j = w(S j) - w j`. An affine weight gives a *nonzero* Christoffel yet *zero* curvature. *Coq.* `InfoDiscreteRiemannCurvature.v`: `christoffel_can_be_nonzero_while_flat`. *Tier.* [Th_coqc]. *Why.* Curvature must be a *second-order* object; a nonzero connection that is still flat proves the first difference alone is gauge, not geometry.

Step 8 — Riemann = second difference. *Statement.* The Riemann curvature tensor [17] is the second finite difference $\text{riemann_fd} = w(S(Sj)) - 2w(Sj) + wj$; affine $\Rightarrow$ flat, and the non-affine witness $w = qn^2$ gives $\text{riemann_fd}0 = 2 \neq 0$; flat $\Leftrightarrow$ locally affine (arithmetic progression). *Coq.* `riemann_is_second_difference`, `affine_is_flat`, `curvature_nonzero_witness`, `flat_iff_second_diff_zero`. *Tier.* [Th_coqc]. *Why.* This is the same second difference that defines L_R (Step 2): geometry and the root operator are the same object read curvature-first.

Step 9 — Riemann = group commutator (Heisenberg, division-free). *Statement.* Genuinely two-index curvature is the group commutator — a matrix-mechanics commutator [3, 11] and, around a plaquette, a discrete Gauss–Bonnet holonomy [8, 2] — on the Heisenberg / upper-unipotent group (polynomial inverse, so *no division*): R_{xy} = the centre of $[X, Y] = a \cdot b$, with $[Y, X] = -a \cdot b$; abelian, scalar, and same-direction plaquettes are all flat. *Coq.* `InfoDiscreteRiemannCommutator.v`: `plaquette_curvature_z`, `reverse_antisymmetric`, `curvature_is_central`. *Tier.* [Th_coqc]. *Why.* Holonomy around a plaquette is a commutator; realising it in a nilpotent *group* keeps everything in $\mathbb{Q}$ and delivers kl-antisymmetry exactly, without any limit or inverse.

Step 10 — ij-antisymmetry + Jacobi = first Bianchi ($\mathfrak{so}(3)$). *Statement.* On the metric-compatible algebra $\mathfrak{so}(3) \cong (\mathbb{Q}^3, \times)$ the generator `asym u` is antisymmetric, the matrix commutator satisfies $[\text{asym } u, \text{asym } w] = \text{asym}(u \times w)$, curvature is ij-antisymmetric, and the *Jacobi identity is the first Bianchi identity* [1]. *Coq.* `InfoMetricCompatibleCurvature.v`: `asym_is_antisymmetric`, `commutator_is_asym_cross`, `curvature_ij_antisymmetric`, `bianchi_jacobi`. *Tier.* [Th_coqc]. *Why.* Metric compatibility forces the connection into $\mathfrak{so}(3)$, where the cross product closes the bracket and the algebraic Bianchi identity is just Jacobi — proven, division-free.

Step 11 — rational $SO(2)/SO(3)$ isometry (Cayley). *Statement.* Genuine rotations exist over $\mathbb{Q}$: $SO(2)$ via the Cayley / Pythagorean map [4] $N = (I - A)\text{adj}(I + A)$ with $N^T N = (1 + t^2)^2 I$; and a rational $SO(3)$ capstone (two (3, 4, 5) rotations, holonomy $\neq I$) carrying nonzero rational curvature. *Coq.* `InfoRationalIsometry.v`, `InfoRationalS03Curvature.v`. *Tier.* [Th_coqc]. *Why.* Transport must be an isometry; the Cayley map gives exact rational orthogonal transport with *no* $\sqrt{\cdot}$, so curved holonomy is a $\mathbb{Q}$ -readout.

Step 12 — metric-derived Levi-Civita (rational, g^{-1} no $\sqrt{g}$). *Statement.* From a metric $g = \text{diag}(1, w)$ the Levi-Civita / Christoffel symbol [5] $\Gamma_{12}^2 = \frac{1}{2} dw/w$ is rational for $w \neq 0$ (it needs only g^{-1} , never $\sqrt{g}$), and $R_{1212} = -\frac{1}{2} ddw + (dw)^2/(4w)$: flat on constant w , curved witness $R_{1212} =$

$-\frac{1}{4}$. *Coq.* `InfoMetricDerivedCurvature.v`: `christoffel_G212_rational`, `flat_constant_zero_curvature`, `curved_witness`. *Tier.* [Th_coqc]. *Why.* The connection is *derived from the metric* (not posited) and remains rational because only the inverse metric enters; the square root that an orthonormal frame would demand is exactly the object refused in Step 14.

Step 13 — second Bianchi $dF = 0$; pair symmetry derived. *Statement.* The abelian differential Bianchi identity [1] holds on the 3D lattice, $d_z F_{xy} + d_x F_{yz} + d_y F_{zx} = 0$, from mixed-difference commutation (nonzero-curvature witness, not vacuous); and the pair symmetry $R_{ijkl} = R_{klij}$ is *derived* from A1 (ij-antisym) + A2 (kl-antisym) + B1 (first Bianchi), for abstract R over $\mathbb{Q}$. *Coq.* `InfoDiscreteSecondBianchi.v`: `second_bianchi`, `mixed_differences_commute`; `InfoRiemannPairSymmetry.v`: `pair_symmetry`. *Tier.* [Th_coqc]. *Why.* With all four Riemann symmetries (kl-antisym, ij-antisym, first Bianchi, pair symmetry) and both Bianchi identities machine-checked, the discrete curvature tensor is *structurally complete* (cpg DEC-curvature-chain-complete).

7 The $\sqrt{g}$ frame refused (I1 non-readout)

Step 14 — where the thread stops, on purpose. *Statement.* An orthonormal frame requires $\sqrt{g}$. For a non-perfect-square metric such as $\text{diag}(1, 2)$ there is *no rational frame*: the frame is an I1 injected infinity ($\mathbb{R}$ -completeness), while the connection and curvature of Step 12 stay rational. *Coq.* `InfoMetricFrameNonReadout.v`: `metric_frame_diag12_no_rational`. *Tier.* [Th_coqc] $\rightarrow$ [Refused]. *Why.* The theory reads the geometry through g^{-1} (rational), and *declines* the frame: $\sqrt{g}$ is a non-readout no agency ever reads, so it is refused as a *position, verified where refused* — not a gap. This is the local face of the Continuum Refusal that replaces any “graph-to-continuum” gate: the discrete tensor PDE over $\mathbb{Q}$ is complete on its own terms.

8 The naive non-abelian Bianchi defect

Step 15 — an honest negative, exactly measured. *Statement.* Extending the differential Bianchi identity to the non-abelian (group) case by the *naive* covariant difference fails, and it fails by an exact amount: the Leibniz defect

$$(\text{naive covariant Bianchi defect}) = \text{cross}(dA, dB).$$

Coq. `InfoDiscreteLeibnizObstruction.v`. *Tier.* [Th_coqc] (an exact identity for the *defect*). *Why.* The finite Leibniz rule carries a cross-term that a genuine

non-abelian Bianchi identity must absorb through transport; measuring the defect exactly is what turns the correct non-abelian statement into a well-posed *positive* target (Step O2), rather than a vague “does it work” question.

9 The honest ledger

Nothing in §§1–8 tagged [Th_coqc] is open: the RD root, L_R , the spine PDE, the telegraph mass, the discriminant crossover, and the entire curvature tensor (Steps 7–13, all four symmetries and both Bianchi identities) are closed, axiom-free results over $\mathbb{Q}$. What remains is cleanly split into *genuine positive extensions* and *refused non-readouts*. The two must not be confused.

[Open] — **genuine positive research targets (not gaps in the closed results).**

[Open] **O1 — full R^i_{jkl} in $n \geq 3$.** The complete metric-derived Riemann array satisfying all symmetries, from a general (non-diagonal, non-affine) metric g — beyond the $\text{diag}(1, w)$ single component R_{1212} of Step 12.

[Open] **O2 — exact non-abelian (group) second Bianchi.** In transport / holonomy-cube form; the naive covariant form provably fails by the exact defect $\text{cross}(dA, dB)$ of Step 15, so the correct statement needs the transport-corrected cube.

[Open] **O3 — general parametrised $SO(n)$ family.** Tied to a metric g ; $n = 2$ ($SO(2)$) and a witnessed rational $SO(3)$ are closed (Step 11), the general family is open.

[Open] **O4 — Schrödinger reduction and absolute constants.** The non-relativistic Schrödinger reduction, and the numeric values of the absolute constants M, D, K, τ_c — OPEN.CONSTANTS: there is no external anchor for these absolute values, and none is claimed. The complex/real *root magnitudes* of the telegraph characteristic are *not* listed here: their *sign/regime* over $\mathbb{Q}$ is already [Th_coqc] (Step 6), and their transcendental *values* are [Refused] (below), not an unfinished target.

[Refused] — **non-readouts declined on thesis grounds (not gaps; §7).**

[Refused] The **continuum** Riemann tensor / smooth d^2g , $d\Gamma$, and any continuum limit of the spine PDE (I2, $h \rightarrow 0$).

[Refused] The **orthonormal frame $\sqrt{g}$** (I1) — proven non-readout for non-perfect-square g (Step 14).

[Refused] **Born-rule uniqueness** (I1).

[Refused] The **real/complex root magnitudes** of the telegraph characteristic — the transcendental phase/rate *values* in +reals (I1 non-readout). The *sign/regime* over $\mathbb{Q}$ is [Th_coqc] (Step 6); the magnitude *values* are declined, not left [Open].

[Refused] Full continuum GR / Einstein dynamical closure *as a target*; the GR content enters only through the discrete $K \cdot L_R$ geometry readout and the horizon knife-edge (Step 6d), both [Th_coqc].

Do not tag any of Steps 2–13 [Open]: the discrete tensorial spine and the curvature chain are closed [Th_coqc]. The absolute constant values are [Open] (no anchor, none claimed); the continuum and $\sqrt{g}$ are [Refused]. The continuum is the map, not the territory.

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Code Availability. The Coq sources of every machine-checked claim, maintained at <https://github.com/morrocwi/causal-quantum-gravity> (formal/, with a `Makefile` and reproduction commands), constitute the artifact and reproduce every machine-checked claim of line; each file compiles under `coqc 8.20.1` with exit status 0 and `Print Assumptions = Closed under the global context`. Every commit is additionally re-verified end-to-end by continuous integration (`make verify` on GitHub Actions); the badge is the authoritative live count.

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